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Palaeomagnetism of some Precambrian rocks in south-east Greenland

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A palaeomagnetic pole is established at 25.1°N 273.9°E (dp = 10.6°, dm = 14.3°) from the norite–charnockite complex at Angmagssalik, emplaced at 1800 Ma. A somewhat older palaeomagnetic pole at 4.2°S 246.7°E (dp = 4.2°, dm = 8.3°) is obtained from Archaean gneisses close to the northern boundary of the Nagssugtoqidian mobile belt; reversals of magnetization are present here. Both magnetizations were imposed during slow cooling following the (late) Nagssugtoqidian metamorphism.

In general the gneisses, dyke amphibolites and granite of the Nagssugtoqidian mobile belt are unstably magnetized; their magnetization is attributable to the Earth's present field, and is often extremely weak.

A pseudotachylyte within the Archaean gneisses has had a long cooling history. A fragment of the remanence reflects the magnetization characteristic of the Archaean gneisses, whereas most of the magnetization corresponds to a palaeomagnetic pole near that of the Angmagssalik complex. The pseudotachylyte is much older than its magnetizations.

An apparent polar wander path is presented for Greenland at ca. 1750 Ma based on the above results and data from west Greenland.

1. Introduction

A palaeomagnetic collection of coastal rocks near Angmagssalik was made with the Geological Survey of Greenland in 1976 (Fig. 1). This investigation is a contribution to a substantial programme of geological research on the Nagssugtoqidian mobile belt of east Greenland (Bridgwater, 1976; Bridgwater et al., 1977, 1978; Bridgwater and Myers, 1979; Pedersen and Bridgwater, 1979).

In east Greenland the Nagssugtoqidian is restricted to a 240 km wide belt of rocks flanked on either side by Archaean gneisses which were only slightly affected by the Nagssugtoqidian metamorphism. This mobile belt was established at about 2600 Ma during a period of intense shearing (Nag 1), with syntectonic emplacement of suites of basic dykes.

The Nagssugtoqidian mobile belt was deformed by a major thermal episode (Nag 2) at 1850 ± 150 Ma (Pedersen and Bridgwater, 1979). An important manifestation of this episode was the intrusion of the norite–charnockite complex, a body measuring at least 70 km × 8 km in surface extent. It is believed to have been emplaced at 1800 Ma, in view of the uranium–lead zircon age determined by R. Chessex (cited in Pedersen and Bridgwater, 1979). The complex was emplaced just before or possibly during the youngest phase of Nagssugtoqidian deformation (Bridgwater and Myers, 1979), i.e., at the end of Nag 2.

Nag 2 was contemporaneous with the (Late) Laxfordian metamorphism in north-west Scotland

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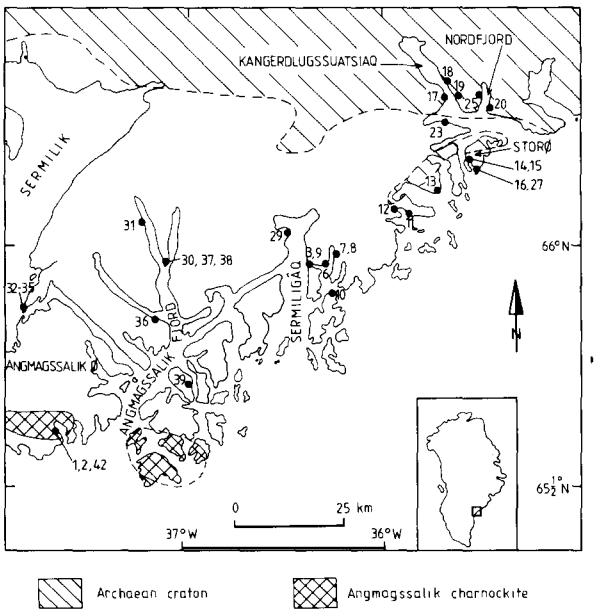


Fig. 1. Map of the sampling area, showing sites discussed in the text. The Angmagssalik charnockite and the northern boundary of the Nagssugtoqidian mobile belt with the Archaean are indicated (after Bridgwater and Myers, 1979). Land areas left white represent the northern part of the mobile belt.

(Kalsbeek et al., 1978). This association was established radiometrically. A parallel correspondence is to be expected between the palaeomagnetic records. Such a link has been proposed (Beckmann, 1979), and formed the basis of reconstructions of the relative palaeopositions of Greenland and north-west Scotland (Piper, 1979; Beckmann, 1981).

The principal objective of the present study is to compare palaeomagnetic results from the Nagssugtoqidian belt and Archaean rocks on the east coast of Greenland with the corresponding data from the west coast (reviewed by Beckmann, 1979; see also Piper, 1981).

The results will be interpreted assuming that the stable magnetizations were established during slow cooling due to uplift (Morgan, 1976). On this model even a single specimen may contain the complete magnetic record of its locality. Cooling is so slow that changes of direction of the Earth's field due to secular variation are averaged out. Thus, a palaeomagnetic pole determined from a single specimen is a true palaeomagnetic pole, and not a virtual geomagnetic pole. In many palaeomagnetic studies, one of the reasons for taking a number of samples at each site is to eliminate secular variation. In studies such as the present one, the reason for taking many samples at each site is purely to obtain good statistical control.

2. Sampling

Palaeomagnetic sampling extended from the Angmagssalik norite–charnockite complex northeastwards through the Nagssugtoqidian belt, where the gneisses were mainly of amphibolite facies, and into the granulite facies rocks of the Archaean craton in the vicinity of Nordfjord (Fig. 1). Altogether 547 separately orientated samples were collected using a portable drill. There were 33 sites, comprising 25 gneiss localities, 6 dyke amphibolites, 1 post-tectonic granite and 1 pseudotachylyte. At 22 sites, orientation was by sun-compass (Stone, 1967), whereas at 11 sites orientation by Brunton Compass was carried out in conjunction with topographic sighting. At many of the sites both methods were used for the sake of comparison. Orientations of the fiducial line on samples, by the two methods, differed by no more than 2°. Incidentally, the local direction of the Earth's magnetic field has Declination = 324°, Inclination = +78°.

3. Laboratory methods

Samples were 2.5 cm diameter, and typically 7 cm long. One standard specimen, 2.5 cm long was cut from the lower end of each sample. The original intention was to treat specimens by thermal demagnetization at 500°C, 540°C, and perhaps higher temperatures, in view of the success of this regimen with metamorphic rocks in west Greenland (Beckmann et al., 1977). However, some of the present specimens became adversely affected by heat (see section 4.1.), and so a more cautious approach was indicated. For many of the sites, a two stage treatment programme was adopted, first

alternating field (AF) demagnetization at relatively low fields, and then thermal demagnetization. This procedure was advocated by Roy and Lapointe (1978) and followed by Briden et al. (1979). This method determines whether the end direction isolated by AF cleaning is really single- or multi-component, and enables better isolation of the original component than by using either technique on its own.

Directions and intensities of magnetization were measured on a Digico spinner magnetometer (Molyneux, 1971) in the Nuffield palaeomagnetic laboratory at Wylam, near Newcastle. Thermal and AF demagnetization were done in this laboratory using locally-made apparatus. Further work, at greater AFs and higher temperatures, was carried out in the palaeomagnetic laboratory of the United States Geological Survey in Jeddah; this is equipped with a Schonstedt spinner magnetometer, and Schonstedt AF and thermal demagnetizers.

Directions of magnetization were analysed using the statistics of Fisher (1953) in which k is a measure of the precision of grouping and α_{95} is the semi-angle of the cone of 95% confidence about the mean direction. Further information about the remanence directions has been obtained by the use of vector subtraction techniques.

4. Palaeomagnetic results

4.1. The Nagssugtoqidian belt

The palaeomagnetic results from rocks within the mobile belt were extremely poor, except for those from the norite–charnockite complex which will be treated separately (section 4.1.1.). The results from 22 sites will be considered in this section. Sites 6, 8, 12, 13, 16 and 30 are dyke amphibolites, site 32 is a granite, whilst the remaining 15 sites are gneisses. The locations of these sites are shown in Fig. 1. Sites 14, 23 and 27 occur close to the boundary with the Archaean craton and their results have affinity with those from that region, and so it is preferred to treat these three sites later (section 4.2.).

No useful palaeomagnetic results have been obtained from the 22 sites. In many cases, specimens demagnetize unduly readily, resulting in highly scattered directions of magnetization, whose mean directions, even when significant at the 95% confidence level, bear no relation to known directions of the Earth's ancient magnetic field, observable in Greenland.

The overwhelming characteristic of the magnetization in these specimens is their closeness to the Earth's recent magnetic field, assumed to be, on average, a simple geocentric axial dipole. On this model, the Earth's field in the sampling region would have Declination (Dec) = 0°, Inclination (Inc) = +77.5°. This will be known as the present Earth's field (PEF) direction. It is believed that practically all the magnetization in the 22 sites is due to the PEF. Analysis of the initial magnetization (NRM), the directions after the first two demagnetizing stages, and the vectors removed, has borne out this belief (Table I). The mean directions listed at the foot of Table I are all very close to the PEF direction. Indeed the cones of 95% confidence about these five mean directions each contain the PEF direction. Thus the significant magnetizations are apparently inseparable from the PEF direction.

The magnetization observed at these sites may be attributed to a viscous magnetization acquired during the (present) Brunhes normal epoch of the Earth's magnetic field. (The only possible exception to this generalization is site 3.) Some of the 22 sites are extremely weakly magnetized. For example, site 32 (the granite) has an average intensity of magnetization of 3×10^{-6} emu cm^{-3}. Sites 33, 34, 36 and 37, have average NRM intensities of 1.5×10^{-6}, 4×10^{-7}, 8×10^{-7} and 4×10^{-6} emu cm^{-3}, respectively. These low intensities required long measuring times on the magnetometers. For this reason it was not feasible to carry out more than one demagnetizing step on specimens from such sites, in view of the large numbers of specimens requiring treatment. At site 29, eight of the specimens were found to have increased their intensities of magnetization, and their magnetic susceptibilities, by very many times after heating to 500°C. Accordingly the statistics at this site are based on 15 specimens instead of the 23 collected. The

TABLE I

Some directions of magnetization from the Nagssugtoqidian belt

Site no.	Demagnetizing fields [b] or temperatures	N	NRM			V_1		
			Dec	Inc	α_{95}	Dec	Inc	α_{95}
3	50, 100, 150 Oe; 540°C	20	54.7	61.8	3.7	53.2	60.6	4.3
6	50, 100, 150, 200 Oe; 540°C	21	332.4	79.4	3.1	336.7	77.2	3.0
7	100, 150 Oe	20	326.1	59.8	20.0	337.2	51.1	24.4
8	50, 100 Oe	12	352.5	70.2	11.0	349.2	78.8	19.7
9	100, 150 Oe; 540°C	21	33.8	75.8	6.8	21.8	76.9	9.4
10	500°C	22	331.4	53.8	10.2	N.S.		
11	100, 150 Oe	21	16.6	75.6	3.2	11.8	73.6	4.3
12	100, 150, 200 Oe; 540°C	21	353.4	75.4	3.9	335.2	70.1	6.5
13	100, 150, 200 Oe; 540°C	20	2.1	78.7	3.3	30.5	78.3	4.6
15	400°C	21	345.4	62.7	15.8	[a]		
16	100 Oe	12	[a]			N.S.		
29	500°C	15	20.7	59.1	19.4	N.S.		
30	100 Oe	12	[a]			N.S.		
31	500°C	12	312.5	60.6	21.4	[a]		
32	50, 100 Oe	12	9.3	70.5	15.4	36.7	73.4	16.0
33	400°C	13	N.S.			N.S.		
34	500°C	11	N.S.			N.S.		
35	50, 100 Oe	11	N.S.			N.S.		
36	500°C	13	N.S.			N.S.		
37	500°C	12	84.7	65.3	18.3	N.S.		
38	500, 540°C	11	86.8	79.1	5.6	276.2	66.6	24.8
39	500, 540°C	12	226.6	85.9	19.3	N.S.		
Mean Directions			2.6 (16 sites)	74.2	7.6	357.8 (10 sites)	74.9	9.5

N = the number of specimens, where each specimen is cut from one independently orientated sample.
V_1 = the site mean direction after the first demagnetizing step.
C_1 = the mean of all the vector differences between the NRM and the first demagnetizing step.
V_2 = the mean direction after the second demagnetizing step.

behaviour indicates the creation of new magnetic material, implying mineralogical instability.

4.1.1. The Angmagssalik charnockite

The norite–charnockite complex at Angmagssalik was sampled in three places. The rocks at sites 1 and 2 are garnetiferous granulite, whilst site 42 is a norite.

The rocks from sites 1 and 2 (0.7 km apart) were of exceptionally high palaeomagnetic quality and will be discussed in detail. Demagnetization at 100 and 150 Oe revealed extremely stable and precise magnetization with site mean directions of magnetization close to each other, suggesting that only a single component of magnetization remained. For site 1, treatment was continued with AF demagnetization, whereas site 2 was treated thermally. If the two treatments failed to separate the two mean directions, we would have powerful proof that the same single component remained in both sites. The results are presented in Table II.

For site 1 high values of k resulted from all stages of demagnetization. The directions of magnetization are extremely steady after 500, 600 and 800 Oe. The vectors removed after each step have also been studied. The mean vector removed in the step 600 to 800 Oe has Declination = 235.2°, Inclination = 59.2° ($k = 21$, $\alpha_{95} = 7.4°$) which is indistinguishable from the direction of the main vector after 600 and 800 Oe. This confirms that

C_1 = "NRM-V_1"			V_2			C_2 = "V_1-V_2"		
Dec	Inc	α_{95}	Dec	Inc	α_{95}	Dec	Inc	α_{95}
57.7	63.9	4.2	50.6	58.6	4.8	56.4	65.7	6.4
329.3	80.7	3.6	326.9	72.3	4.2	351.6	80.9	4.4
351.3	79.9	24.0	328.7	49.3	26.1	[a]		
354.4	67.8	12.4	[a]			287.0	77.5	22.0
48.9	77.2	8.3	15.3	75.4	11.9	47.3	78.9	19.5
21.2	77.3	3.4	2.9	71.9	5.2	36.0	80.6	11.9
5.7	77.0	5.0	327.2	65.9	11.5	356.4	77.1	17.3
347.0	78.0	3.8	50.7	76.7	6.5	331.4	77.0	8.0
[a]			13.6	68.7	24.7	N.S.		
N.S.								
N.S.								
N.S.			N.S.			N.S.		
N.S.								
88.7	75.9	5.1	[a]			[a]		
			[a]			N.S.		
22.1 (9 sites)	78.0	6.8	0.5 (8 sites)	70.8	11.1	11.4 (7 sites)	80.4	8.3

C_2 = the mean of all the vector differences between the first and second demagnetizing steps.
[a] = $\alpha_{95} > 30°$. The mean direction is considered too imprecise to tabulate.
N.S. = the mean direction has no significance at the 95% confidence level (Irving, 1964, p. 63).
[b] unit of demagnetizing field is (peak) oersted (Oe).

TABLE II

Palaeomagnetic results for the garnetiferous granulite at Angmagssalik

Demagnetizing field (Oe)	Site 1 (20 Specimens)						Site 2 (21 Specimens)						
	Magnetization		k	α_{95}	Palaeomagnetic pole		Demagnetizing field or temp.	Magnetization		k	α_{95}	Palaeomagnetic pole	
	Dec	Inc			°N	°E		Dec	Inc			°N	°E
0	268.9	76.1	61	4.2	54.4	272.8	0 Oe	274.6	80.2	9	11.0	60.6	280.7
100	246.9	65.0	156	2.6	33.7	273.4	100 Oe	235.4	60.0	396	1.6	24.8	279.1
150	240.3	62.7	277	2.0	29.1	276.8	150 Oe	233.9	57.5	311	1.8	21.8	279.2
400	236.6	59.6	534	1.4	24.7	278.0	500°C	235.7	55.1	319	1.8	19.9	276.8
500	234.8	59.0	692	1.2	23.5	279.2	540°C	249.3	59.6	5	15.8	28.7	268.2
600	234.9	59.5	565	1.4	24.1	279.3	550°C	244.7	63.3	2	35.4	31.1	273.9
800	235.0	59.5	331	1.8	24.1	279.2							
1000	230.6	59.1	238	2.1	22.5	282.5							

the main magnetization at this site has only one component. Thus, the direction after 600 Oe is adopted as the cleaned direction of magnetization. Even after 1000 Oe (the highest field available) k is still large; moreover one third of the initial intensity of magnetization remains (Fig. 2(a)). The shape of the demagnetization curve shows that the remanence resides in magnetite; the high coercivities are attributable to single domain grains (Evans and McElhinny, 1969).

Site 2 yielded large values of k after AF demagnetization at 100 and 150 Oe and after thermal treatment at 500°C. The change of mean direction between 150 Oe and 500°C was only 2.7° whilst the intensity fell only slightly (Fig. 2(b)). This similarity of response to the two treatments is proof of magnetic reliability. The direction of magnetization after 500°C is adopted as the cleaned direction. Subsequent thermal cleaning at 540°C caused the intensity of five of the specimens to drop to about 1% of their initial intensity, with concomitant scattering of their directions. Nevertheless five other specimens are exceptionally stable to thermal demagnetization (Fig. 2(c)); in the five steps between 500° and 580°C, their average movement of mean direction per step was < 1°. After 580°C, $k = 302$, but after 590°C, $k = 3$. This sudden scatter, accompanied by the sharp fall of intensity close to the Curie point of magnetite, demonstrates again that in the garnetiferous granulite, the remanence resides in magnetite.

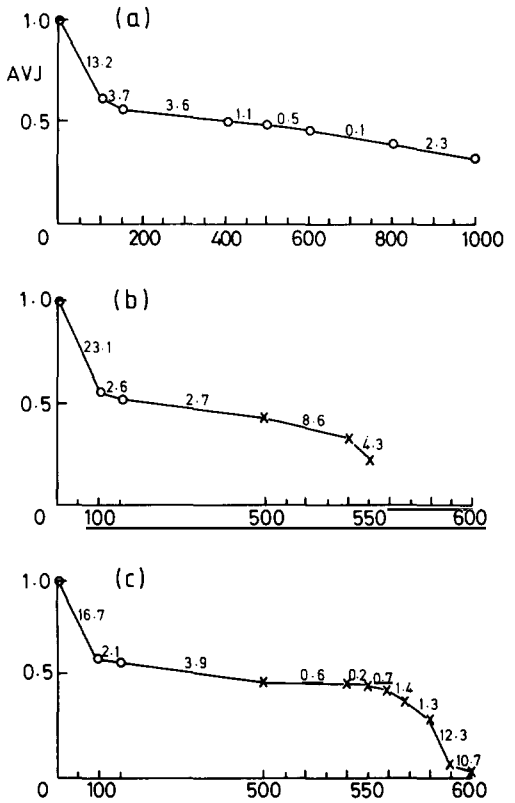


Fig. 2. Demagnetization curves for sites 1 and 2. AVJ is the average value of J/J_0 where J is the intensity of magnetization of any specimen after a particular treatment, and J_0 its initial intensity: (a) = site 1; (b) = site 2, all specimens; (c) site 2, picking the five very stable specimens; o = AF demagnetizing stage; x = thermal demagnetizing stage. Units of horizontal axes are Oe and °C as appropriate. Numbers between points represent the angles in degrees moved by the mean vector in a treatment step (ANG).

The norite specimens (site 42) were treated at 500, 540, 550, 560, 570, 580, 590, 600 and 620°C. The magnetization at this site is less clear. The best grouping of directions is achieved after 540°C with $k = 120$. However, further movement of 18° in the mean direction of magnetization occurs in the treatment step 540–550°C. After 560°C, $k = 88$ and the mean vector moved only 1° in the step 550–560°C, and so the direction after 560°C is adopted as the optimum direction of magnetization at this site, corresponding to a palaeomagnetic pole at 31.4°N 264.9°E. Nevertheless, this stationary position is only temporary, because after higher treatment temperatures the mean pole wanders south-eastwards with ever decreasing precision. After 580°C, $k = 14$, and the pole position is at 18.9°N 281.6°E, very close to the palaeomagnetic poles derived from sites 1 and 2.

A summary of the results from Angmagssalik is presented in Table III.

4.2. The Nordfjord gneisses

Stable magnetization has been discovered in some of the sites on Storø and northwards in the Archaean craton in the neighbourhood of Nordfjord and Kangerdlugssuatsiaq (Fig. 1), but it is necessary to be selective with the data. Sites 14, 17, 18, 19, 20 and 23 are gneisses of granulite facies, 27 is a Nag 1 shear zone in the gneisses, whilst 25

TABLE III

Optimum palaeomagnetic results from Angmagssalik (65.6°N 37.6°W)

Site no.	N	Treatment	Magnetization		k	α_{95}	Palaeomagnetic		dp	dm
			Dec	Inc			°N	°E		
1	20	600 Oe	234.9	59.5	565	1.4	24.1	279.3	1.5	2.1
2	21	500°C	235.7	55.1	319	1.8	19.9	276.8	1.8	2.5
42	10	560°C	254.3	60.6	88	5.2	31.4	264.9	6.0	7.9
Mean: 3 sites			241.2	58.7	164	9.6	25.1	273.9	10.6	14.3

dp and dm are the semi-axes of the ovals of 95% confidence about the mean palaeomagnetic pole.

is a pseudotachylyte (see section 4.3).

The gneisses have been treated by thermal demagnetization, generally at 500°C and 540°C, and at higher temperatures. For three sites thermal demagnetization was preceded by AF demagnetization at 100 and 150 Oe. It became clear that the quality of the palaeomagnetic results, as judged by the smallness of the angles turned by the magnetic vector in a demagnetizing step, and consistency of directions between sites, was associated with the average relative intensity of specimens after treatment (AVJ). The palaeomagnetic quality of the specimens represented by the two upper curves in Fig. 3 is much better than for those in the curves beneath. Measurements of NRM and those after treatment are set out in Table IV.

Site 20, with 21 specimens, has two groups of directions. Eighteen of the specimens (Group 1) have directions of magnetization after cleaning which are nearly horizontal and towards the east-north-east, with a corresponding palaeomagnetic pole near the equator at about longitude 70°E. This group of specimens is approximately reversed with respect to the remaining three specimens from site 20 (Group 2). The theory of magnetization during slow cooling requires that the Earth's magnetic field is predominantly of one sense during the long time taken for a rock to cool through its critical range of blocking temperatures. The usual consequence of this proviso is that samples from a site all have the same polarity. However, if there are distinct differences in coercivity between samples, those with the higher coercivities may have become magnetized earlier, and it may be that at such times the field was predominantly of the opposite sense from that prevailing when the lower coercivity samples were magnetized. Such a mechanism could have operated in the present case

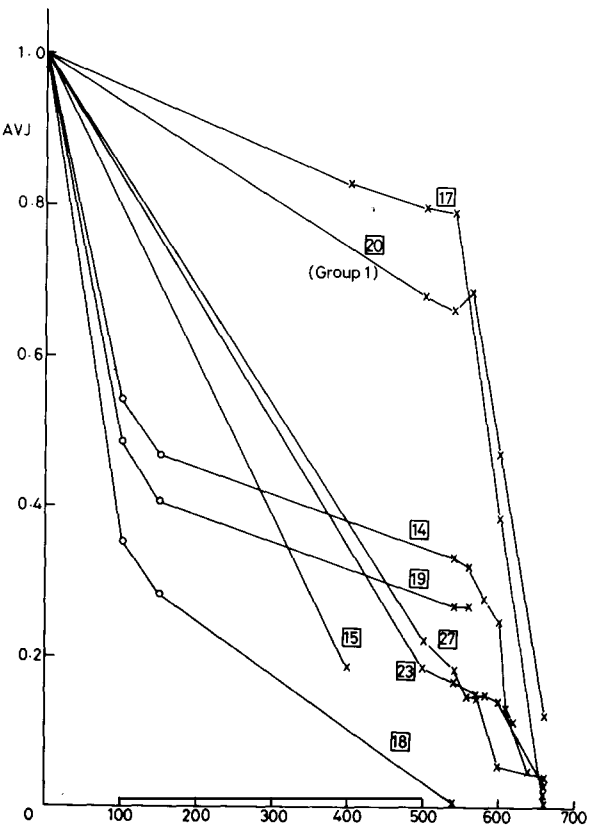


Fig. 3. Relative intensity vs. treatment (as in Fig. 2) for Nordfjord gneisses (and site 15). Site numbers in squares.

TABLE IV

Palaeomagnetic results for the Nordfjord gneisses: initially and after 540°C

Site no.	N	NRM directions				Data after treatment at 540°C									
						Magnetization				Palaeomagnetic pole				ANG	AVJ
		Dec	Inc	k	α_{95}	Dec	Inc	k	α_{95}	°N	°E	dp	dm		
17	21	259.2	23.0	8	11.9	252.2	1.8	22	6.9	−6.2	251.0	3.5	6.9	1	0.79
20 (Group 1)	18	53.0	58.0	8	12.8	72.7	−4.3	24	7.2	4.9	71.3	3.6	7.2	6	0.66
20 (Group 2)	3	278.5	70.0	8	46.2	270.6	33.5	22	26.7	16.9	241.5	17.3	30.4	3	0.44
14	21	297.9	61.5	11	9.9	248.3	4.2	5	15.4	−6.6	255.3	7.7	15.5	4	0.34
19	21	295.7	56.5	17	8.0	273.9	6.9	4	18.1	4.7	232.2	9.1	18.2	9	0.27
27	6 †	296.7	63.3	41	10.6	260.6	13.5	6	29.4	2.5	245.9	15.3	30.0	3	0.21
23	11	321.9	79.4	20	10.6	59.8	32.8	7	18.8	28.2	75.2	12.0	21.2	10	0.17
18	15	352.3	69.6	22	7.4	67.4	0.9	7	15.4	9.3	75.0	7.7	15.4	68	0.09

ANG = the angle turned by the mean vector in the previous demagnetizing step.
† shear zone

because the relative intensities of the Group 1 specimens after treatment are 50% greater, implying higher blocking temperatures, than those in Group 2. Thus, the Group 1 magnetization may be older than that established in Group 2.

In Table IV it is apparent that after treatment, the mean directions from sites 18, 23 and 20 (Group 1) are approximately reversed with respect to the other sites. This phenomenon could be explained by differences in stratigraphic height of the sites, resulting in cooling through the critical range of blocking temperatures at different times, associated with differences in the predominant magnetic field polarity.

The data in Table IV are not ideal for defining a mean palaeomagnetic pole for the Nordfjord region on account of the large values of α_{95} at some of the sites, and the fact that the group of poles is not compact. However, the observation that the better results are associated with the two upper curves in Fig. 3 suggests a method of enhancing the data. All those specimens have been picked for which $J/J_0 > 0.3$ after 540°C; the results are shown in Table V (in which the site 20 groups are combined by reversing two of the Group 2 specimens). Sites 23 and 27 have only 1 specimen satisfying the criterion whilst site 18 has none. The criterion has resulted in smaller values of α_{95} at sites 17, 20, 14 and 19. There is good between-site grouping of directions for these sites, leading to a well-defined mean direction: Dec = 257.0°, Inc = 2.0° ($k = 121$, $\alpha_{95} = 8.3$) and a corresponding palaeomagnetic pole at 4.2°S, 246.7°E.

The data on which the above result is based are thoroughly reliable. Site 19 is typical and bears out this claim (Table VI). The mean direction and intensity of magnetization are steady between 540° and 590°C. Thereafter the group begins to scatter; after 620°C the mean direction has no significance at the 95% confidence level. The behaviour of four stable specimens from site 20 is illustrated in Fig. 4. Three of these maintain their intensity practically unaltered to at least 600°C whereas specimen 8 shows a sharp fall of intensity between 560°C and 600°C. In site 17 many of the specimens retain their initial intensity after treatment at 600°C, whereas in others the intensity falls from the initial value to about 10% in the step 540°–600°C. Magnetization appears to reside in magnetite and haematite at these sites, but no difference in directional behaviour can be seen between these two phases.

4.3 The pseudotactylyte

This dyke-like body (site 25) in Nordfjord is a few cm wide. It is cut by a basic dyke, 1.2 m wide, whose age is either Jurassic (Mitchell, 1978) or

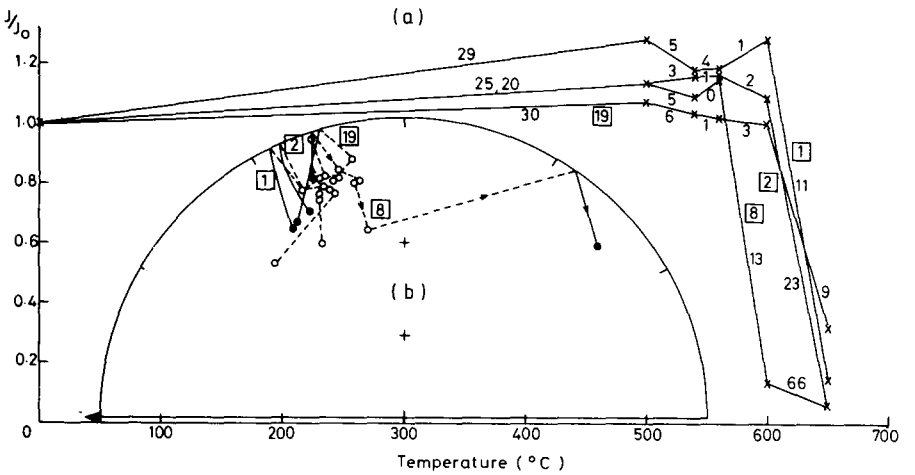


Fig. 4. Behaviour of four stably magnetized specimens from site 20 during thermal demagnetization: (a) = relative intensity of specimens 1, 2, 8, 19; specimen numbers in squares. Other numbers are angles turned in demagnetizing steps. (b) = stereographic projection of corresponding directional changes. ● (○) = positive (negative) inclination.

TABLE V

Palaeomagnetic results for the Nordfjord gneiss specimens having $J/J_0 > 0.3$ after 540°C

Site no.	N	Treatment (°C)	Directions of magnetization		k	α_{95}	Palaeomagnetic pole		dp	dm
			Dec	Inc			°N	°E		
17	19	540	254.0	−1.7	46	5.0	−7.1	248.6	2.5	5.0
20	17	540	74.2	−9.6	31	6.5	−1.8	250.9	3.3	6.6
14	8	540	255.0	−0.4	41	8.7	−6.1	248.1	4.4	8.7
19	8	540	264.9	0.5	31	10.1	−1.8	239.1	5.1	10.1
Mean: 4 sites			257.0	2.0	121	8.3	−4.2	246.7	4.2	8.3

Tertiary (Beckmann, 1982). At the time when the pseudotachylyte was sampled there was no knowledge about its age, except that it was pre-Tertiary. Ten samples were taken of the pseudotachylyte at distances from the dyke varying from 0.3 m to 4.3 m. A contact test on the dyke using palaeomagnetic results from the pseudotachylyte has already been described (Beckmann, 1982). No original remanence can be found in the specimen nearest the dyke, and so only the remaining nine specimens will be considered (from 0.9 to 4.3 m from the dyke).

The specimens were treated at 50 and 100 Oe and then at 500, 540, 560, 580, 590 and 600°C (Table VII). Figure 5(a) illustrates variation of the mean values of relative intensity (AVJ) during treatment; clearly the remanence resides in magnetite.

The mean direction remains rather steady during treatment whilst k improves, reaching a maximum value of 23 after 560°C. After 580°C it has fallen to 4; thereafter the directions scatter. After 540°C the direction of magnetization in each specimen lies close to that of the corresponding vector subtracted in the step 540°–560°C (Fig. 5(b)). This point is seen again in the mean direction after 540°C; this direction is indistinguishable from that of the mean vector subtracted in the step 540–560°C (see Table VII). Thus, after 540°C, the magnetic vector consists primarily of one component. Nevertheless, the distribution of directions does not conform to that of Fisher (1953) (Fig. 5(b)).

TABLE VI

Detailed behaviour of the group of 8 specimens selected from site 19 (Table V)

Treatment	Direction of magnetization		k	α_{95}	Palaeomagnetic pole		ANG	AVJ
	Dec	Inc			°N	°E		
NRM	282.9	44.4	17	13.8	28.9	233.9		1.00
100 Oe	268.3	22.0	18	13.4	9.8	240.5	25	0.66
150 Oe	268.2	15.6	23	11.8	6.6	239.1	6	0.63
540°C	264.9	0.5	31	10.1	−1.8	239.1	15	0.51
560°C	263.0	1.2	35	9.6	−2.2	240.9	2	0.47
580°C	257.3	−0.4	21	12.4	−5.2	245.9	6	0.45
590°C	258.3	−1.8	30	10.3	−5.5	244.7	2	0.45
600°C	255.3	6.7	10	18.8	−2.8	249.2	9	0.37
610°C	252.0	20.0	6	24.4	2.4	254.8	14	0.19
620°C	N.S.							0.17

TABLE VII

Palaeomagnetic results from the pseudotachylyte (site 25): a row under a demagnetizing stage represents the mean vector subtracted in the previous step

Treatment	N	Direction of magnetization		k	α_{95}	°N	°E	ANG	AVJ
		Dec	Inc						
NRM	9	265.6	52.2	7.8	19.6	28.0	252.8		1.00
50 Oe	9	264.1	48.8	7.2	20.5	24.7	252.5	3	0.82
	9	291.0	75.8	11.8	15.6	61.9	261.1		
100 Oe	9	265.1	50.2	9.3	17.8	26.2	252.3	1	0.71
	9	N.S.							
500°C	9	264.8	46.1	18.9	12.2	23.0	250.8	4	0.57
	9	N.S.							
540°C	9	270.1	48.0	21.6	11.3	26.5	247.0	4	0.50
	9	238.5	27.5	4.4	27.8	1.6	268.8		
560°C	9	265.1	49.1	23.2	10.9	25.4	251.8	3	0.27
	9	271.2	45.4	14.6	14.0	24.9	245.0		
580°C	9	216.8	−1.7	3.9	29.8	−19.6	285.0	66	0.13
	9	313.3	57.7	9.8	17.4	51.7	211.4		
590°C	8	248.9	33.3	2.1	50.8	8.5	260.8	46	0.02
	8	216.2	−8.8	3.7	33.4	−23.2	284.7		
600°C	5	N.S.							0.02
	5	N.S.							

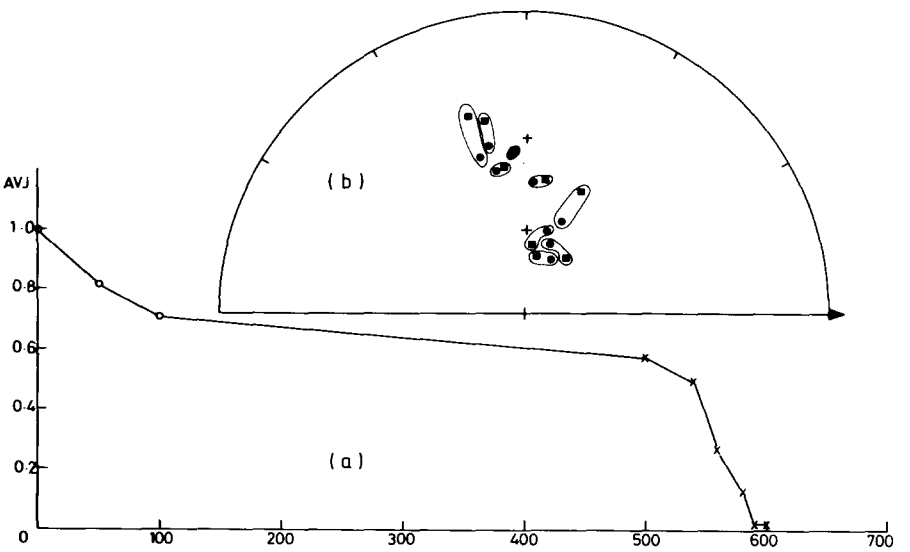


Fig. 5. Behaviour of pseudotachylyte specimens (site 25) during demagnetization: (a) Average relative intensity (AVJ) vs. treatment, symbols as in Fig. 2; (b) = Directions of magnetization for individual specimens after 540°C (●); (■) = directions of vectors removed in treatment step 540–560°C.

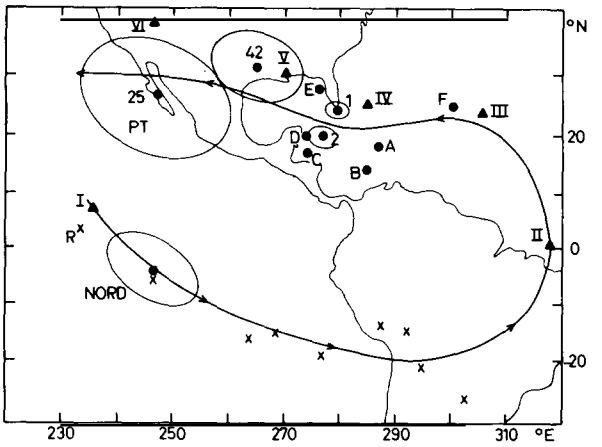


Fig. 6. APWP for Greenland at ca. 1750 Ma. Palaeomagnetic poles derived in this study are points surrounded by ovals of 95% confidence. NORD = Nordfjord mean pole based on four sites. 1, 2 and 42 are Angmagssalik gneisses, 1800 U [1]. PT = pseudotachylyte after 540°C; crosses represent poles from individual specimens after 580°C; the point R is a reversed pole. A = Kangâmiut dykes in Itivdleq, 1750 K [2]. B = Kangâmiut dykes and gneisses in Itivdleq [3]. C = Kangâmiut dykes, mainly in Archaean craton [4], 1950R [5]. D = Godthåb gneisses [4], 1600–1700 K, R [6]. E = Nordre Strømfjord gneisses [7], 1710 K [8], 1800 R [9]. F = Kællingehætten gneisses [7]. Poles I–VI (▲) from Piper (1981). Radiometric ages in Ma are denoted by numbers (above) followed by K for K–Ar, R for Rb–Sr, or U for the U–Pb (zircon) method. References: [1] = R. Chessex in Pedersen and Bridgwater (1979); [2] = Beckmann and Mitchell (1976); [3] = Morgan (1976); [4] = Fahrig and Bridgwater (1976); [5] = Kalsbeek et al. (1978); [6] = Pankhurst et al. (1973); [7] = Beckmann et al. (1977); [8] = Larsen and Møller (1968); [9] = Hickman (1979).

After 580°C, immediately before the remanence vanishes (at 590°C), the individual poles from the nine specimens have the distribution shown by crosses in Fig. 6. Thus, the directions after 540°C defined only a temporarily stationary position of the palaeomagnetic pole at 26.5°N 247.0°E (dp = 9.7°, dm = 14.8°).

5. Discussion

The palaeomagnetic poles derived in this study are displayed in Fig. 6. The Angmagssalik charnockite poles are assigned an age of 1800 Ma in view of the emplacement of this body at about this time. A palaeomagnetic pole in this position and with this age accords with results from west Greenland (Fig. 6). The ideal behaviour of the garnetiferous granulite sites (1 and 2) is reminiscent of that of the Kangâmiut dykes, whose magnetization was established during slow cooling (Morgan, 1976).

The magnetic stability of the Angmagssalik complex is in dramatic contrast to the ubiquitous unstable magnetization in the Nagssugtoqidian belt to the north of it. Evidently Nag 2 events degraded the quality of the magnetic minerals in those rocks so that they were unable to record the Nag 2 cooling history. Indeed they are unable to record the Earth's magnetic field for longer than the present normal epoch.

According to the geological evidence, the Angmagssalik complex was emplaced near the end of Nag 2. This probably occurred after the destructive events just mentioned, so that the complex was able to record the end of Nag 2 cooling.

Only minor structural effects were imposed by Nag 2 on the Archaean gneisses in the Nordfjord area. Nevertheless, these rocks were at metamorphic temperatures at this time (Bridgwater and Myers, 1979). Thus the palaeomagnetic pole from the Nordfjord gneisses may also be associated with cooling following Nag 2, because this was the last metamorphic event to have affected these rocks.

The palaeomagnetic pole obtained from the Nordfjord gneisses differs from that obtained from Angmagssalik in two ways.

(1) They lie far apart (Fig. 6). The distinction between the two poles becomes even clearer when the palaeolatitude of southern Greenland is considered. These are 39° and 1°, derived from the Angmagssalik and Nordfjord inclinations of magnetization, respectively.

(2) The Nordfjord gneisses have mixed polarity, both within sites and between sites. Greenlandic samples yielding palaeomagnetic poles in the region of the West Indies all have the same polarity (Beckmann, 1979); this generalization is strongly reinforced by new data (Piper, 1981).

Thus, the Nordfjord gneisses have a distinctly different magnetic age from the Angmagssalik charnockite, but which is older? As the Angmagssalik complex was emplaced at the close of Nag 2,

it is reasonable to suggest that its magnetization occurred later than that observed in the Nordfjord gneisses.

Very interestingly, Piper (1981) reported palaeomagnetic poles from late Nagssugtoqidian diorites in west Greenland (Praestefjeldet section, marked as pole I in Fig. 6) which lie close to those of the Nordfjord gneisses and, like those gneisses, display bipolarity. These late Nag 2 diorites have been dated at 1700 Ma by the Rb–Sr method (Hickman, 1979). Both regions display high coercivity magnetization, possibly residing partly in haemo-ilmenite. On the slow cooling theory of magnetization (Morgan, 1976), high coercivity magnetic material will become magnetized earlier than lower coercivity material and thus record an earlier section of the apparent polar wander path (APWP). Piper (1981) made extensive use of this principle; in particular, he used it to suggest that the Praestefjeldet section is older than the poles lying near the West Indies. He proposed that the connexion is via an anticlockwise loop passing through northern South America. This proposal was based on a similar trend discerned in palaeomagnetic poles from North American rocks of similar age. The results of this paper, regarding the argument of relative ages of the magnetizations, support the APWP of Piper (1981) who preferred to present the Greenland data as if they were North American data by making a correction for Phanerozoic drift of Greenland away from North America. However, there are now enough data for an early Precambrian APWP for Greenland to stand on its own.

A major reservation about applying the proposition that haematite will be magnetized earlier than magnetite during slow cooling is that the haematite might be secondary. However, the fact that a similar palaeomagnetic pole and reversals have been obtained from Praestefjeldet and Nordfjord, in spite of their separation of ca. 700 km, is striking evidence that genuine palaeomagnetic poles, associated with a primary haematitic phase, are being observed.

The palaeomagnetic pole from the pseudotachylyte lies quite close to that of the Angmagssalik gneisses (Fig. 6), indicating that final cooling of both occurred at similar times. However, the highest coercivity grains have given palaeomagnetic poles falling on the APWP from the Nordfjord gneiss pole; there is even a reversed pole from one of the specimens (marked R in Fig. 6). Thus, this last fragment of the remanence represents an earlier phase of cooling history in the pseudotachylyte and simultaneously supports the southerly part of the loop suggested by Piper (1981).

Pseudotachylytes are formed in country rocks which are nearly cold, say 150°C (Sibson, 1975). In these circumstances the palaeomagnetic data would have indicated a rapid cooling history. In contrast, the long cooling history observed in this pseudotachylyte suggests that it is a much earlier structure (than say 1750 Ma) which was subsequently reheated and then cooled with the adjacent country rocks, probably by deep burial and uplift. This possibly happened in Archaean times.

An early Precambrian APWP is proposed for Greenland (Fig. 6) based on the APWP of Piper (1981), taking due account of all the data. This path was established during cooling following Nag 2. It has not proved possible to correlate the APWP with any trend in isotopic ages. Perhaps the most suitable estimate of the age of the West Indies section of the path is that of Beckmann and Mitchell (1976). In that study K–Ar whole rock ages were obtained for three Kangâmiut dykes, and K–Ar isochrons derived for two gneiss sites, all in Itivdleq, west Greenland. A regional cooling age of about 1750 Ma was suggested; this was envisaged as the time at which the K–Ar isotope system closed and the magnetization became blocked during regional uplift following the Nagssugtoqidian metamorphism (Nag 2).

Piper (1981) suggested that, after his pole VI, the APWP follows a north-north-westerly course to 68°N 238°E. It is difficult to decide whether such northerly poles are true palaeomagnetic poles or reflect a component arising from the PEF. In this study, the component removed in the first demagnetizing step or steps often appeared to arise from the PEF. This is true even for the very stable sites 1 and 2 in which this effect would be least expected. For example, in site 1 the directions of the mean vectors removed in the first three demagnetizing steps are: Dec = 357.9°, Inc = 75.9° ($\alpha_{95} = 9.0°$); Dec = 323.0°, Inc = 74.3°

($\alpha_{95} = 13.9°$); Dec = 316.0°, Inc = 75.1° ($\alpha_{95} = 9.2°$). These directions lie close to the PEF direction and the magnetic field direction actually observable in the field (section 2). The third of the directions just cited would correspond to a palaeomagnetic pole at 70.6°N 242.2°E.

For a Kangâmiut dyke with an AF demagnetizing curve similar to these Angmagssalik granulites, Morgan and Smith (1981) showed that there were two distinct types of remanence carrier; those carriers with unblocking fields < 150 Oe were of multidomain type residing in large magnetite grains, whereas those with unblocking fields 200–1200 Oe were essentially of single domain type residing in submicron magnetite within the plagioclase. It is well-known that single domain grains are capable of carrying magnetic remanence for very much longer times than multi-domain grains. Thus, these experiments provide strong grounds for believing that the Kangâmiut dykes and the garnetiferous granulite at Angmagssalik furnish oustandingly reliable records of the Earth's early Proterozoic magnetic field.

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References

Beckmann, G.E.J., 1979. A review of Nagssugtoqidian palaeomagnetism. Rapp. Grønlands Geol. Unders., 89: 115–124.

Beckmann, G.E.J., 1981. Comments on "The palaeomagnetism of the Central Zone of the Lewisian foreland, north-west Scotland" by J.D.A. Piper. Geophys. J. R. Astron. Soc., 66: 463–470.

Beckmann, G.E.J., 1982. Palaeomagnetism of nine dated Phanerozoic dykes in south-east Greenland. Geophys. J. R. Astron. Soc., 69: 355–367.

Beckmann, G.E.J. and Mitchell, J.G., 1976. Palaeomagnetic and geochronological work in central west Greenland. Earth Planet. Sci. Lett., 30: 269–280.

Beckmann, G.E.J., Olesen, N.Ø. and Sørensen, K., 1977. A palaeomagnetic experiment on crustal uplift in west Greenland. Earth Planet. Sci. Lett., 36: 269–279.

Briden, J.C., Duff, B.A. and Kröner, A., 1979. Palaeomagnetism of the Koras Group, Northern Cape Province, South Africa. Precambrian Res., 10: 43–57.

Bridgwater, D., 1976. Nagssugtoqidian mobile belt in East Greenland. In: A. Escher and W.S. Watt (Editors), Geology of Greenland. Geol. Surv. Greenland, Copenhagen, pp. 97–103.

Bridgwater, D., Davies, F.B., Gill, R.C.O., Gorman, B.E., Henriksen, N. and Watterson, J., 1977. Field mapping in the Nagssugtoqidian of south-east Greenland. Rapp. Grønlands Geol. Unders., 85: 74–83.

Bridgwater, D., Davies, F.B., Gorman, R.C.O., Myers, J.S., Pedersen, S. and Taylor, P., 1978. Precambrian and Tertiary geology between Kangerdlugssuaq and Angmagssalik, south-eastern Greenland. Rapp. Grønlands Geol. Unders., 83: 17 pp.

Bridgwater, D. and Myers, J.S., 1979. Outline of the Nagssugtoqidian mobile belt of east Greenland, Rapp. Grønlands Geol. Unders. 89: 9–18.

Evans, M.E. and McElhinny, M.W., 1969. An investigation of the origin of stable remanence in magnetite-bearing igneous rocks. J. Geomagn. Geoelectr., 21: 757–773.

Fahrig, W.F. and Bridgwater, D., 1976. Late Archaean—Early Proterozoic palaeomagnetic pole positions from west Greenland. In: B.F. Windley (Editor), The Early History of the Earth, Wiley, London, pp. 427–439.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London, Ser. A., 217, 295–307.

Hickman, M.H., 1979. A Rb–Sr age and isotope study of the Ikertoq, Nordre Strømfjord, and Evighedsfjord shear belts, west Greenland—outline and preliminary results. Rapp. Grønlands Geol. Unders., 89: 125–128.

Irving, E., 1964. Palaeomagnetism and its Application to Geological and Geophysical Problems. Wiley, New York, 399 pp.

Kalsbeek, F., Bridgwater, D. and Zeck, H.P., 1978. A 1950 ± 60 Ma Rb–Sr whole-rock isochron age from two Kangâmiut dykes and the timing of the Nagssugtoqidian (Hudsonian) orogeny in West Greenland. Can. J. Earth Sci., 15: 1122–1128.

Larsen, O. and Møller, J., 1968. Potassium–Argon age studies in west Greenland. Can. J. Earth Sci., 5: 683–691.

Mitchell, J.G., 1978. Potassium–argon ages from Phanerozoic basic dykes in south-east Greenland. Rapp. Grønlands Geol. Unders., 90: 141–145.

Molyneux, L., 1971. A complete result magnetometer for measuring the remanent magnetization of rocks. Geophys. J. R. Astron. Soc., 24: 429–433.

Morgan, G.E., 1976. Palaeomagnetism of a slowly cooled plutonic terrain in western Greenland. Nature, 259: 382–385.

Morgan, G.E. and Smith, P.P.K., 1981. Transmission electron microscope and rock magnetic investigations of remanence carriers in a Precambrian metadolerite. Earth Planet. Sci. Lett., 53: 226–240.

Pankhurst, R.J., Moorbath, S., Rex, D.C. and Turner, G., 1973. Mineral age patterns in ca. 3700 m.y. old rocks from west Greenland. Earth Planet. Sci. Lett., 20: 157–170.

Pedersen, S. and Bridgwater, D., 1979. Isotopic re-equilibration of Rb–Sr whole rocks systems during reworking of Archaean gneisses in the Nagssugtoqidian mobile belt, east Greenland. Rapp. Grønlands Geol. Unders., 89: 133–146.

Piper, J.D.A., 1979. The palaeomagnetism of the central zone of the Lewisian foreland, north-west Scotland. Geophys. J. R. Astron. Soc., 59: 101–122.

Piper, J.D.A., 1981. The altitude dependence of magnetic remanence in the slowly-cooled plutonic terrain of west Greenland. Earth Planet. Sci. Lett., 54: 449–466.

Roy, J.L. and Lapointe, P.L., 1978. Multiphase magnetizations: problems and implications. Phys. Earth Planet. Int., 16: 20–37.

Sibson, R.H., 1975. Generation of pseudotachylyte by ancient seismic faulting. Geophys. J. R. Astron. Soc., 43: 775–794.

Stone, D.B.., 1967. A sun compass for the direct determination of geographic North. J. Sci. Instrum., 44: 661–662.